Class - 11th
Chapter - Complex Numbers And Quadratic Equations
Hand Written Notes

Class -11th

Complex Numbers

$$Z = \underbrace{x}_{\text{Real part } Re(z)} + \underbrace{iy}_{\text{Imaginary part } Im(z)}$$

$i = \sqrt{-1}$
$i^2 = -1$
$i^3 = -i$
$i^4 = 1$

$Z = 2 + 3i$
$\downarrow$ $\quad\downarrow$
$Re(z) \quad Im(z)$

Also, $i^{1728} = (i^4)^{432} = 1$

$i^{1727} = (i^4)^{431}(i)^3 = -i$

Ques: $z = \left(\frac{1}{4-3i}\right)$ $\quad Re(z) = ?$ $\quad Im(z) = ?$

Solu: $z = \left(\frac{1}{4-3i}\right)\left(\frac{4+3i}{4+3i}\right) = \frac{4+3i}{16+9} = \frac{4}{25} + \frac{3}{25}i$

$\frac{4}{25}$ ↓ $Re(z)$ $\quad$ $\frac{3}{25}$ ↓ $Im(z)$

Points to Remember...

1. If $A, B \in R$ then $z = A + iB$ is a complex no., but if $z = A + iB$ is a complex no. then it is not necessary that $A, B \in R$

Example:- $z = (2+3i) + (2+4i)i$ → is a complex no.

but A, B are not real

2. If imaginary part of complex no. is zero, then z is known as purely real

Example:- $z = 4$

3. If real part of complex no. is zero, then z is purely imaginary

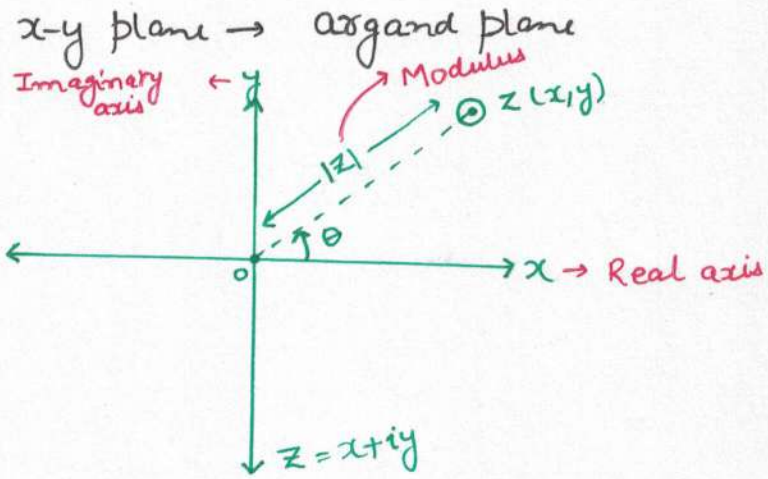
Example:- $z = 4i$

4. If $B \neq 0$, then z is imaginary number

5. A complex no. which is imaginary ($B \neq 0$) is non-comparable

6. In an argand plane, the complex no. is a point, where argand plane is nothing but a co-ordinate axes,

$x \rightarrow$ real axis $y \rightarrow$ imaginary axis



7. **Modulus of z** : The distance between the origin and point z in argand plane is called magnitude or modulus of z.

$$|z| = \sqrt{x^2+y^2}$$

OR

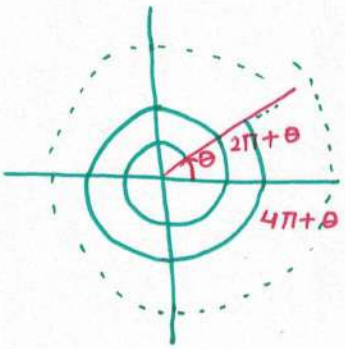
$$|z| = \sqrt{(Re(z))^2 + (Im(z))^2}$$

8. **Argument of z** : The angle made by the line segment with +ve direction of x-axis in anticlockwise is known as argument of z.

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{Im(z)}{Re(z)}\right)$$

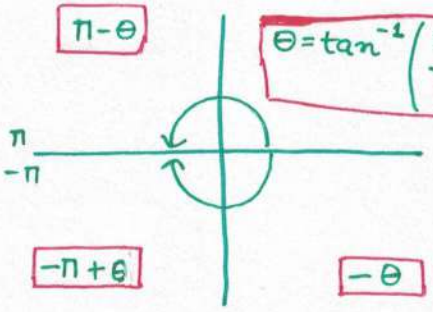
$\arg(z) = \theta$

General argument | **Principle argument** | **Lead +ve argument**



∞ arguments

$\arg(z) = \theta + 2n\pi \quad \{n \in I\}$



$amp(z) = \theta \quad \theta \in [-\pi, \pi]$

$\arg(z) = \theta \quad \theta \in [0, 2\pi]$

☞ arg (z) behaves like log while principle argument amp (z) does not behave like log

Ques: $z = -3 - 3\sqrt{3}\,i$

$\arg(z) = \tan^{-1}(\sqrt{3}) = \pi/3$

gen. $\arg(z) = \frac{4\pi}{3} + 2n\pi$

$amp(z) = -2\pi/3$

Least +ve $\arg(z) = \frac{4\pi}{3}$

Ques: Find the $\arg(z) = ?$ $\quad z = \dfrac{(1+i)^7\,(\sqrt{3}+i)^9}{(1+\sqrt{3}\,i)^5}$

$\arg(z) = 7\arg(1+i) + 9\arg(\sqrt{3}+i) - 5\arg(1+\sqrt{3}i)$

$= 7(\pi/4) + 9(\pi/6) - 5(\pi/3)$

Arg (z) behaves like log

$$\arg(z^n) = n \arg(z)$$
$$\arg(z_1 z_2) = \arg z_1 + \arg z_2$$
$$\arg(z_1 / z_2) = \arg z_1 + \arg z_2$$

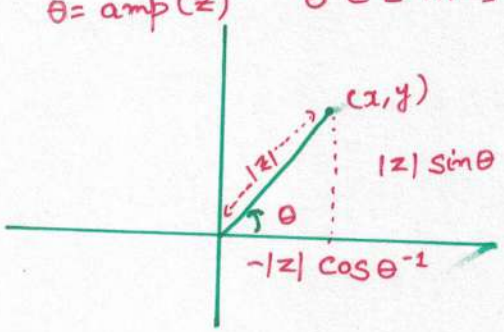
DIFFERENT FORMS OF COMPLEX NUMBERS

1 Co-Ordinal Form

$$z = x + iy \qquad [x, y \in R]$$

2 Polar Form

$\theta = \text{amp}(z)$ $\quad \theta \in [-\pi, \pi]$



$$z = |z| \cos\theta + i|z| \sin\theta$$
$$z = r\cos\theta + ir\sin\theta$$
$$z = r(\cos\theta + i\sin\theta)$$
$$z = r(\text{cis}\,\theta)$$

3 EULAR'S Form

$$z = r(e^{i\theta}) \qquad e^{i\theta} = \text{cis}\,\theta$$

eg: $z^n = r^n (e^{i\theta})^n = r^n (e^{i\theta n})$

$$\arg(z^n) = n \arg(z)$$

eg: $z_1 z_2 = r_1 r_2 e^{i\theta_1} \cdot e^{i\theta_2}$

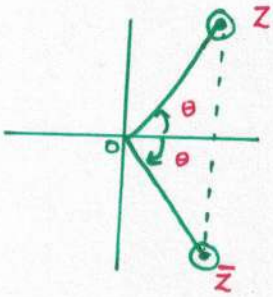
$$= r_1 r_2 e^{i\theta_1 + i\theta_2}$$

$$\arg(z_1 z_2) = \theta_1 + \theta_2$$

$$= \arg(z_1) + \arg(z_2)$$

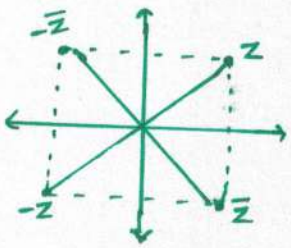
4 Conjugate of z

If $z = x + iy$, then $\bar{z} = x - iy$



NOTES

1. $z + \bar{z} = 2\,Re(z)$
2. $z - \bar{z} = 2i\,Im(z)$
3. $\bar{z} = z \Rightarrow$ z is purely real
4. $\bar{z} = -z \Rightarrow$ z is purely imaginary
5. $z \cdot \bar{z} = |z|^2 = x^2 + y^2$
6. $z, \bar{z}, -z, -\bar{z}$ forms a rectangle



7. Two complex no.'s are said to be equal if their corresponding real parts as well as imaginary parts are equal.

   i.e. z_1 and z_2 are said to be equal if,

   $|z_1| = |z_2|$ as well as $\arg(z_1) = \arg(z_2)$

8. If $z_1 = x_1 + iy_1$ $\quad z_2 = x_2 + iy_2$, then

$$z_1 + z_2 = x_1 + x_2 + i(y_1 + y_2)$$

$$z_1 - z_2 = x_1 - x_2 + i(y_1 - y_2)$$

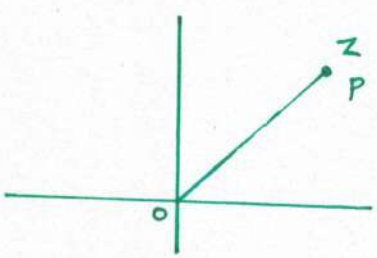
$$z_1 \cdot z_2 = x_1x_2 - y_1y_2 + i(x_1y_2 + x_2y_1)$$

$$\frac{z_1}{z_2} = \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i\,\frac{(x_2y_1) - x_1y_2}{x_2^2 + y_2^2}$$

$$\frac{z_1}{z_2} \times \frac{\bar{z}_2}{\bar{z}_2} = \frac{(x_1 + iy_1)(x_2 - iy_2)}{x_2^2 + y_2^2}$$

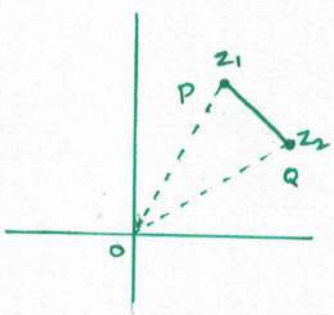
$$= \frac{x_1 x_2 + y_1 y_2 + i(x_2 y_1 - x_1 y_2)}{x_2^2 + y_2^2}$$

9. Complex no. in argand plane can be treated as vector.



$\overrightarrow{OP} = z$

$|z| = |\overrightarrow{OP}|$



$\overrightarrow{PQ} = z_2 - z_1$

$|PQ| = |z_1 - z_2|$

Here, $|z_1 - z_2|$ represents the length of line segment joining

$$|\overrightarrow{PQ}| = |z_2 - z_1| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Ques: Find the locus of z, which satisfies the relation: $|z-2| = |z-4|$

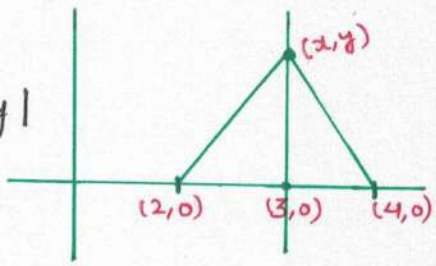
Solution: if $z = x + iy$

$$|(x-2) + iy| = |(x-4) + iy|$$

$$(x-2)^2 + y^2 = (x-4)^2 + y^2$$

$$x - 2 = x - 4$$

$$2x = 6 \Rightarrow \boxed{x = 3}$$



NOTE $|z - z_1| = |z - z_2| \Rightarrow$ z is the perpendicular bisector joining the line segment of z_1, z_2

Ques: Find the locus of z, which satisfy $|z-(2+3i)|=4$

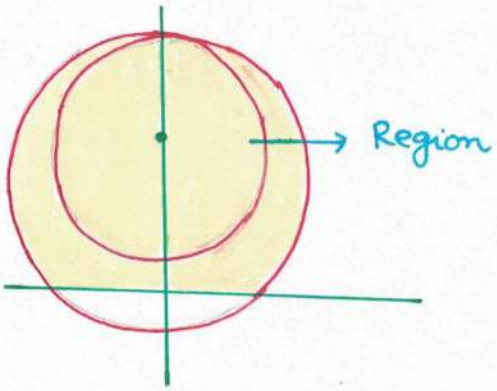
Solution: $|(x-2)+i(y-3)| = 4$

$(x-2)^2 + (y-3)^2 = 16$ → Circle

NOTE $|z-z_1| = a$ } z is the locus of circle with centre z_1 and radius a

Ques: Find the region satisfying $|z-2i|<3$ and $Im(z)\geq 0$

Solution:

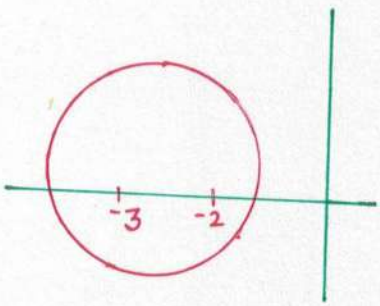


$x^2 + (y-2)^2 < 3$

Ques: $|z+3-i| \geq -2$, find z

Solution: $\sqrt{(x+3)^2 + (y-1)^2} \geq -2$ always true

Whole argand plane



Ques: Find the locus of z, if $\arg(z) = \pi/4$

Solution:

$z = x + iy$

$1 = y/x$

$\Rightarrow \quad x = y$ → Straight Line

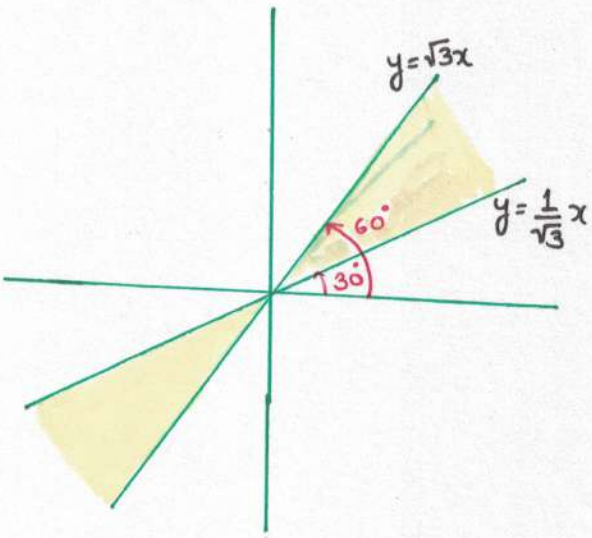
Ques: a) Find the area of the region bounded by-

$$\frac{\pi}{6} \leq \arg(z) \leq \frac{\pi}{3}$$

$z = x + iy$

$\frac{1}{\sqrt{3}} x \leq y$

$\frac{1}{\sqrt{3}} \leq \frac{y}{x} \leq \sqrt{3}$



$$\text{area} = 2\int_0^{\sqrt{3}} x\,dx - \int_0^{\frac{1}{\sqrt{3}}} x\,dx$$

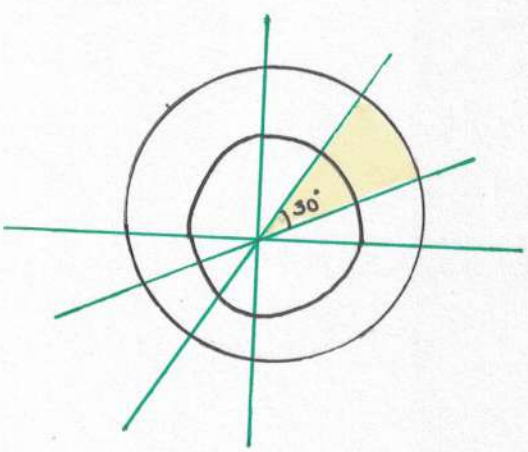
$$= \left[\frac{x^2}{2}\right]_0^{\sqrt{3}} - \left[\frac{x^2}{2}\right]_0^{\frac{1}{\sqrt{3}}}$$

B) Find the area $1 < |z| \leq 3$

$$\text{area} = \frac{1}{2} \times r\theta$$

$$= \frac{1}{2} \times \frac{\pi}{6} \times 4 \times 2$$

$$= \frac{2\pi}{3}$$



Ques: Find the complex no. having least +ve argument and satisfying the inequality $|z-3+3i| \leq 3$

solution

$$\sqrt{(x-3) + i(y+3)} \leq 3$$

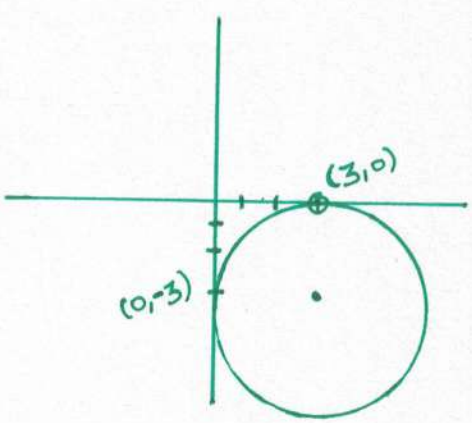
$$(x-3)^2 + (y+3)^2 \leq 9$$

$$(x-3)^2 + 9 = 9$$

$$x = 3 \Rightarrow \boxed{z = 3}$$

$z = x + iy$

$\theta = \tan^{-1}\frac{y}{x}$

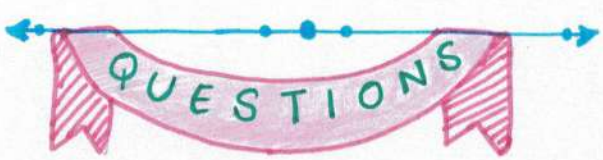


PROPERTIES OF $\bar{z}$

- $\overline{z_1 \pm z_2} = \bar{z}_1 + \bar{z}_2$
- $\overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2$
- $\overline{\frac{z_1}{z_2}} = \frac{\bar{z}_1}{\bar{z}_2}$
- $\overline{z^n} = (\bar{z})^n$

PROPERTIES OF $|z|$

- $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$
- $|z^n| = |z|^n$
- $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$
- $|z| = 1 \Rightarrow z\bar{z} = 1 \Rightarrow \bar{z} = \frac{1}{z}$
- $|z_1 + z_2| \leq |z_1| + |z_2|$
- $||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$ ⟵ Triangle Inequality



Question $|z - i| = 1$ $\arg(z) = \frac{\pi}{2}$ $z = ?$

Solution

$|z| = x + i(y-1)$

$z = (y-1)i$

$z = 2i$

$\frac{y}{x} = \tan\left(\frac{\pi}{2}\right)$ $x \neq 0$

$y - 1 = 1$

$y = 2$

Question $|z + 3 - i| = 4$ $\arg(z) = \pi/4$

Solution

$(x+3)^2 + (y-1)^2 = 16$

$(y-1)^2 = 7$

$y = \sqrt{7} + 1$

$(x+3)^2 + (x-1)^2 = 16$

$2x^2 + 9 + 1 + 6x - 2x = 16$

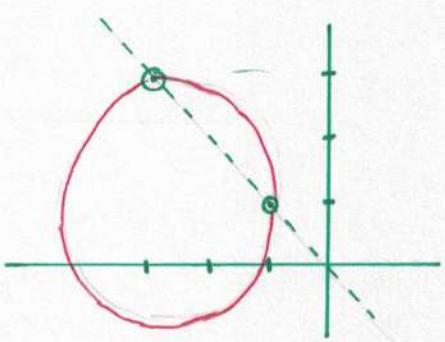
$2x^2 + 4x - 6 = 0$

$x^2 + 2x - 3 = 0$

$x(x+3) - 1(x+3) = 0$

$x = 0$ $x = 1$

$x = -3$



Ques: Solve for z :

a) $z^2 - iz - 1 = 0$

$\Rightarrow \quad z = \dfrac{i \pm \sqrt{i^2+4}}{2} = \dfrac{i \pm \sqrt{3}}{2}$

$\Rightarrow \quad z\bar{z} - i\bar{z} - 1 = 0$

$\Rightarrow \quad \bar{z}(z-i) - 1 = 0$

$\Rightarrow \quad \bar{z}(z-i) = 1$

$z = x+iy$
$\bar{z} = x-iy$

$\Rightarrow \quad (x-iy)[x+i(y-1)] = 1$

$\Rightarrow \quad x^2 + y(y-1) + i$

$\Rightarrow \quad (x+iy)(x-iy) - i(z) = 1$

$\Rightarrow \quad x^2 - y^2 + i(xy + xy) - i(z) = 1$

$\Rightarrow \quad x^2 - y^2 - 1 = i(z - 2xy)$

$\Rightarrow \quad x^2 - y^2 + i\,2xy = 1 + i(z)$

$\Rightarrow \quad 2xy = 2$

$xy = 1$

$x^2 - y^2 = 1$
$x^2 = 1 + y^2$

$\Rightarrow \quad [\sqrt{1+y^2}]\, y = 1$

$\Rightarrow \quad (1+y^2)\, y^2 = 1$

$y^2 = t$

$\Rightarrow \quad t + t^2 = 1$

$\Rightarrow \quad t^2 + t - 1 = 0 \quad \Rightarrow \quad t = \dfrac{-1 \pm \sqrt{5}}{2}$

$\Rightarrow (x+iy)^2 - (x-iy)i - 1 = 0$

$\Rightarrow (x^2 - y^2 + 2ixy) - (x - iy)i - 1 = 0$

$\Rightarrow (x^2 - y^2 - y - 1) + i(2xy - z) = 0 + i0$

$\Rightarrow \quad x^2 - y^2 - y - 1 = 0$

$\Rightarrow \quad x = \pm \dfrac{\sqrt{7}}{2}$

$2xy - x = 0$
$x(2y-1) = 0$
$y = 1/2, \; x = 0$

b) $iz^3 + z^2 - z + i = 0$

$\Rightarrow \quad z(iz^2 + z - 1) + i = 0$

$\Rightarrow \quad i(x^2 - y^2 + 2ixy)(x + iy) = 0$

$\Rightarrow \quad \text{sum} = \frac{-1}{i} \qquad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{-1}{i}$

$\alpha\beta\gamma = \frac{-i}{i} = -1$

$\Downarrow$

$\alpha\beta = \frac{-1}{\gamma}$

$\Rightarrow \quad \alpha + \beta + \gamma = \frac{-1}{i}$

$\gamma(\alpha + \beta) = \frac{-1}{i} + \frac{1}{\gamma}$

$\Rightarrow \quad \gamma = \left(\frac{-1}{i}\right) - (\alpha + \beta)$

$\left[\frac{-1}{i} - (\alpha + \beta)\right](\alpha + \beta) = \frac{-1}{i} + \frac{1}{\gamma}$

$z = i$ is a root

$-\frac{(\alpha + \beta)}{i} - (\alpha + \beta)^2 = 0$

$\Rightarrow \quad i^4 + i^2 - i + i = 0$

$(z - i)(iz^2 - 1) = 0$

$\Rightarrow$

$z - i \;\big)\; iz^3 + z^2 - z + i \;\big(\; iz^2 - 1$

$-iz^3 + z^2$

$-z + 1$

$z^2 = \frac{1}{i} \Rightarrow z = \pm\sqrt{-i}$

$z = \frac{1 \pm i}{\sqrt{2}}$

$\Rightarrow \quad z = x + iy$

$\Rightarrow \quad x^2 + y^2 + 2xyi = -i$

$\Rightarrow \quad x^2 = y^2 \qquad xy = -1/2$

$\Rightarrow \quad x = -y \qquad x = \frac{1}{\sqrt{2}}$

c) Find $|z|$ and amp(z) if $z = 1 + \cos\left(\frac{6\pi}{5}\right) + i\sin\left(\frac{6\pi}{5}\right)$

$|z| = \sqrt{\left(1 + \cos\frac{6\pi}{5}\right)^2 + \left(\sin\frac{6\pi}{5}\right)^2}$

$= \sqrt{2}\sqrt{\left(1 + \cos\frac{6\pi}{5}\right)}$

$= 2\left|\cos\frac{3\pi}{5}\right|$

$= \frac{3\pi}{5}$

$\text{amp}(z) = \frac{2\sin\left(\frac{3\pi}{5}\right)\cos\left(\frac{3\pi}{5}\right)}{2\cos^2\frac{3\pi}{5}}$

$\Rightarrow \theta = \frac{3\pi}{5}$

Ques:- Find the magnitude and arg (z) ⇒ $\left[\frac{(8+i)(1+i)}{5-i}\right]^6$

$$\arg(z) = 6\arg(8+i) + 6\arg(1+i) - 6\arg(5-i)$$

$$= 6\tan^{-1}(1/8) + 6\tan^{-1}(1) + 6\tan^{-1}(1/5)$$

$$= 6\left[\tan^{-1}\frac{\frac{1}{8}+\frac{1}{5}}{1-\frac{1}{40}}\right] + 6\frac{\pi}{4}$$

$$= 6\tan^{-1}\frac{1/3}{39}\ \frac{1}{3}$$

$$= 6\tan^{-1}\frac{\left(\frac{1}{3}+1\right)}{\left(1-\frac{1}{3}\right)}$$

$$= 6\tan^{-1}(4/2) \Rightarrow$$

$$\arg(z) = 6\tan^{-1}(2)$$

$$|z| = \left[\frac{8-1+i(8+1)}{5-i}\right]^6$$

$$= \left[\left(\frac{7+9i}{25+1}\right)(5+i)\right]^6 = \left[\frac{35+9+i(7+45)}{26}\right]^6$$

$$= \left[1+\frac{i(52)}{26}\right]^6 = (\sqrt{5})^6 \Rightarrow |z| = 125$$

Ques:- Let $z_1, z_2, \ldots z_n$ are complex no. in such a way that $z_1, z_2, \ldots z_n$ lying on a circle of center (0,0) and radius = 1.

If $W = \sum_{k=1}^{n}(z_k)\cdot\sum_{k=1}^{n}\frac{1}{z_k}$. Then prove that ⇒

W is real and $W \in (0, n^2)$

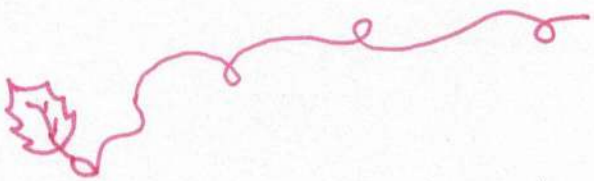
Hint

$|z_1| = |z_2| = \ldots |z_n| = 1$ & $\bar{z}_1 = 1/z_1$

then, $W = (z_1+z_2+\ldots z_n)\cdot\left(\frac{1}{z_1}+\frac{1}{z_2}+\ldots\frac{1}{z_n}\right)$

$$= (z_1+z_2+\ldots z_n)\cdot(\bar{z}_1+\bar{z}_2+\bar{z}_3+\ldots+\bar{z}_n)$$

$= (z_1 + z_2 + \dots z_n)\ (\overline{z_1 + z_2 + \dots z_n})$



Ques: If $|z| \le 1$ and $|w| \le 1$ then prove that :-

$$|z-w|^2 = (|z| - |w|)^2 + \{\arg(z) - \arg(w)\}^2$$



Ques: $\left|z - \frac{4}{z}\right| = 2$, Find max & min value of $|z|$

Solution:

$$\left| |z| - \left|\frac{-4}{z}\right| \right| \le \left| z + \left(\frac{-4}{z}\right) \right| \le |z| + \left|\frac{-4}{z}\right|$$

$$\left| |z| - \frac{4}{|z|} \right| \le 2 \le |z| + \frac{4}{|z|}$$

① $-2 \le |z| - \frac{4}{|z|} \le 2$

$t^2 + 2t - 4 \ge 0$

$t = \frac{-2 \pm \sqrt{4+16}}{2} = \frac{-2 \pm 2\sqrt{5}}{2} = -1 \pm \sqrt{5}$

$= (-\infty, -1-\sqrt{5}) \cup (-1+\sqrt{5}, \infty)$

$t^2 - 4 - 2t \le 0$

$t = \frac{2 \pm \sqrt{4+16}}{2} = 1 \pm \sqrt{5}$

$= [1-\sqrt{5}, 1+\sqrt{5}]$

$t \in [\sqrt{5}-1, \sqrt{5}+1]$

② $t + \frac{4}{t} \ge 2$

$|z| \in [1, \sqrt{5}, 1+\sqrt{5}]$

$t_{max} = 1+\sqrt{5}$

$t_{min} = \sqrt{5}-1$

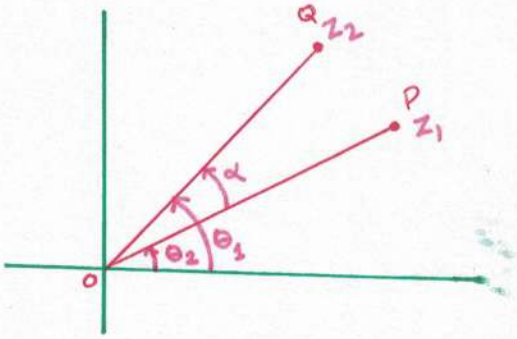
AM, GM

$\frac{t + \frac{4}{t}}{2} \ge \sqrt{4}$

$t + \frac{4}{t} \ge 4$

$\Rightarrow t > 0$

ROTATION OF COMPLEX NO'S



$\alpha = \theta_2 - \theta_1$

$|OP| = |OQ|$

$|z_1| = |z_2| = r$

$z_1 = re^{i\theta_1}$

$z_2 = re^{i\theta_2}$

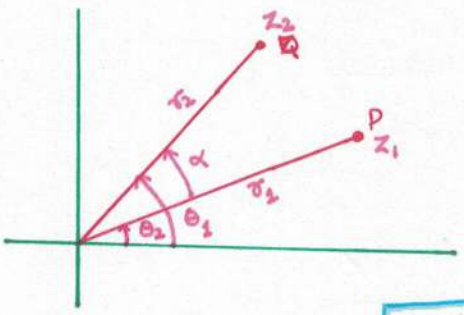
$$\alpha = \theta_2 - \theta_1 = \arg(z_2) - \arg(z_1) = \arg\left(\frac{z_2}{z_1}\right)$$

$$z_2 = re^{i\theta_2} = re^{i(\alpha+\theta_1)} = re^{i\alpha} \cdot e^{i\theta_1}$$

$$\boxed{z_2 = z_1 \cdot e^{i\alpha}}$$

where $e^{i\alpha} = \text{cis}\,\alpha$, $\alpha = \arg(z_2/z_1)$

ROTATION OF COMPLEX NUMBER



$z_2 = r_2 e^{i(\theta_1+\alpha)}$

$z_2 = r_2 e^{i\theta_1} \cdot e^{i\alpha}$

$z_2 = |z_2| \cdot \frac{|z_1|}{|z_1|} e^{i\theta_1} \cdot e^{i\alpha}$

$$\boxed{z_2 = \frac{|z_2|}{|z_1|} \cdot z_1 \cdot e^{i\alpha}}$$

① $|z_1 + z_2| \le |z_1| + |z_2|$

$|z_1 + z_2 + \dots z_n| \le |z_1| + |z_2| + \dots |z_n|$

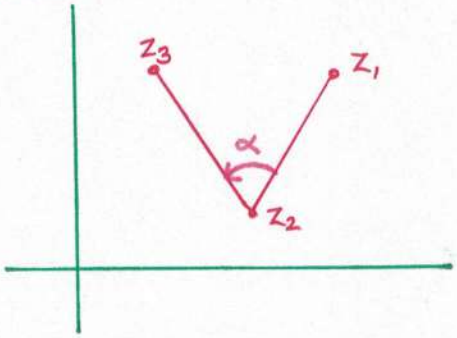
$0 \le |z_1 + z_2 + \dots z_n| \le n$

$0 \le \sqrt{w} \le n$

$0 \le w \le n^2$

② $z_1 = r_1\,\text{cis}\,\theta_1$ $\qquad z_2 = r_2\,\text{cis}\,\theta_2$

ROTATION OF A COMPLEX NO. Z_1 W.R.T. Z_2



$$|Z_1 - Z_2| = |Z_3 - Z_1|$$

$$Z_{final} = Z_{initial}\, e^{i\alpha}$$

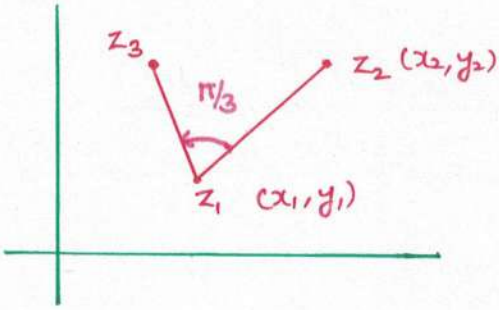
$$Z_3 = Z_2 + (Z_1 - Z_2)e^{i\alpha}$$

$$\boxed{\text{final} = (\text{initial})(e^{i\alpha})}$$

ROTATION OF A COMPLEX NO. Z_1 W.R.T. Z_2 IF MAGNITUDE IS CHANGED

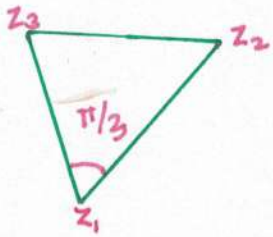
$$\boxed{Z_3 - Z_2 = \frac{|Z_3 - Z_2|}{|Z_1 - Z_2|}(Z_1 - Z_2)\, e^{i\alpha}}$$

TO FIND THE THIRD VERTEX OF AN EQUILATERAL Δ



$$\boxed{\begin{aligned} Z_3 - Z_1 &= (Z_2 - Z_1)\, e^{i\pi/3} \\ Z_3 &= Z_1 + (Z_2 - Z_1)\, e^{i\pi/3} \end{aligned}}$$

Ques: If Z_1, Z_2, Z_3 are 3 verticles of an equilateral Δ, then prove that $Z_1^2 + Z_2^2 + Z_3^2 = Z_1Z_2 + Z_2Z_3 + Z_1Z_3$



Solution:

$$Z_2 - Z_3 = (Z_1 - Z_3)e^{i\theta}$$

$$\Rightarrow \frac{Z_2 - Z_3}{Z_1 - Z_3} = e^{i\pi/3}$$

Also, $\dfrac{Z_1 - Z_2}{Z_3 - Z_2} = e^{i\pi/3}$

$$\Rightarrow \frac{Z_2 - Z_3}{Z_1 - Z_3} = \frac{Z_1 - Z_2}{Z_3 - Z_2}$$

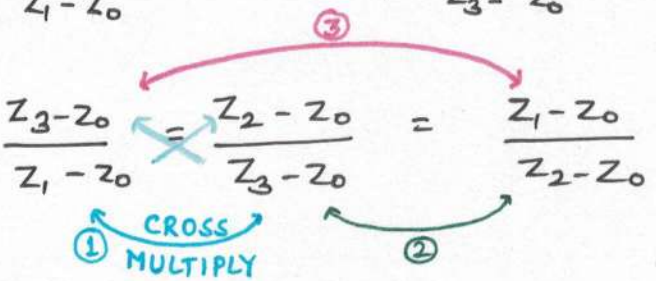
$$\Rightarrow Z_2Z_3 - Z_2^2 - Z_3^2 + Z_2Z_3 = Z_1^2 - Z_1Z_2 - Z_1Z_3 + Z_2Z_3$$

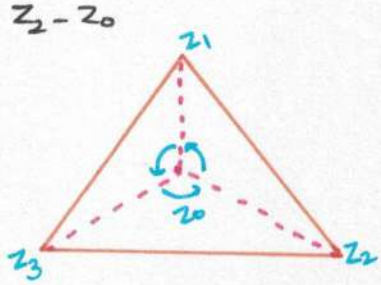
$$\Rightarrow \boxed{Z_1^2 + Z_2^2 + Z_3^2 = Z_1Z_2 + Z_2Z_3 + Z_1Z_3}$$

HENCE PROVED...

Ques: If z_1, z_2, z_3 are the vertex of an eq. Δ, and z tobe centroid, then prove that:- $z_1^2 + z_2^2 + z_3^2 = 3z_0^2$

solu:- $\frac{z_3 - z_0}{z_1 - z_0} = e^{i\theta}$, $\frac{z_2 - z_0}{z_3 - z_0} = e^{i\theta}$, $\frac{z_1 - z_0}{z_2 - z_0} = e^{i\theta}$





$\Rightarrow z_3^2 - z_3z_0 - z_3z_0 + z_0^2 = z_1z_2 - z_1z_0 - z_2z_0 + z_0^2$

$\Rightarrow z_3^2 - 2z_3z_0 = z_1z_2 - 2z_1z_0$ (1)

Similarly $z_2^2 - 2z_2z_0 = z_2z_3 - 2z_2z_0$ (2)

and, $z_1^2 - 2z_1z_0 = z_3z_1 - 2z_3z_0$ (3)

$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$ ✓ EQUATION (A)

If $z_0 = \frac{z_1 + z_2 + z_3}{3} \Rightarrow 3z_0 = z_1 + z_2 + z_3$

$\Rightarrow 9z_0^2 = z_1^2 + z_2^2 + z_3^2 + 2z_1z_2 + 2z_2z_3 + 2z_1z_3$

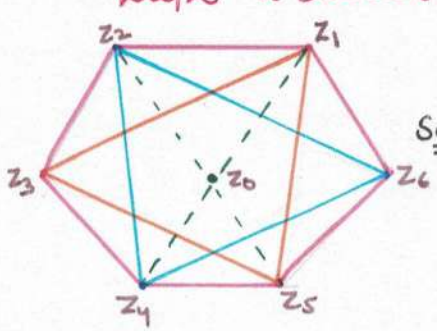
$\Rightarrow 9z_0^2 = z_1^2 + z_2^2 + z_3^2 + 2(z_1z_2 + z_2z_3 + z_3z_1)$

Refer Equation (A)

$\Rightarrow 9z_0^2 = 3(z_1^2 + z_2^2 + z_3^2)$

$\Rightarrow$ **$z_1^2 + z_2^2 + z_3^2 = 3z_0^2$**

Ques: If $z_1, z_2, z_3 \dots z_6$ are vertex of a regular hexagon with a center z_0. Prove that -



$z_1^2 + z_2^2 + z_3^2 + z_4^2 + z_5^2 + z_6^2 = 6z_0^2$

So 2 Eq. Δs : $z_1^2 + z_3^2 + z_5^2 = 3z_0^2$

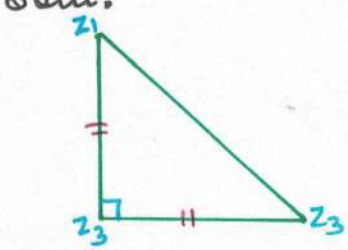
$+ \quad z_2^2 + z_4^2 + z_6^2 = 3z_0^2$

$z_1^2 + z_2^2 + z_3^2 + z_4^2 + z_5^2 + z_6^2 = 6z_0^2$

Ques: If z_1, z_2, z_3 are vertex of a right angle isascles Δ where z_2 has right angle. Prove that:

$$z_1^2 + 2z_2^2 + z_3^2 = 2z_2(z_1 + z_3)$$

Solu:



$$|z_3 - z_2| = |z_1 - z_2|$$

$$\Rightarrow z_1 - z_2 = (z_3 - z_2)(\cos 90^\circ + i \sin 90^\circ)$$

$$\Rightarrow z_1 - z_2 = (z_3 - z_2)(i)$$

$$\Rightarrow z_1^2 + z_2^2 + 2z_1 z_2 = -(z_3^2 + z_2^2 - 2z_3 z_2)$$

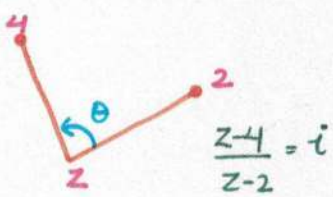
$$\Rightarrow \boxed{z_1^2 + z_3^2 + 2z_2^2 = 2z_2(z_1 + z_3)}$$

Ques: Find the locus of z if,

(a) $\arg\left(\frac{z-4}{z-2}\right) = \frac{\pi}{2}$ (b) $\arg\left(\frac{z-4}{z-2}\right) = \frac{\pi}{4}$

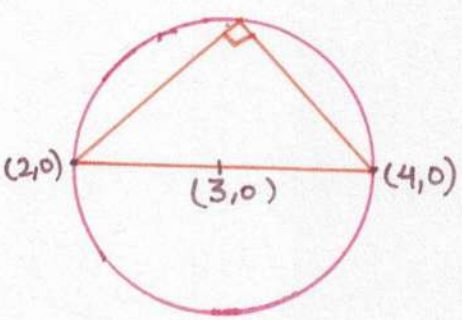
Solu:

(a) $\arg\left(\frac{(x-4)+iy}{(x-2)+iy}\right) = \frac{\pi}{2}$



$$\Rightarrow \arg(z-4) - \arg(z-2) = \pi/2$$

$$\Rightarrow \tan^{-1}\left(\frac{y}{x-4}\right) - \tan^{-1}\left(\frac{y}{x-2}\right) = \pi/2$$



$$\Rightarrow \left(\frac{\frac{y}{x-4} - \frac{y}{x-2}}{1 + \frac{y^2}{(x-4)(x-2)}}\right)$$

$$\Rightarrow (x-4)(x-2) + y^2 = 0$$

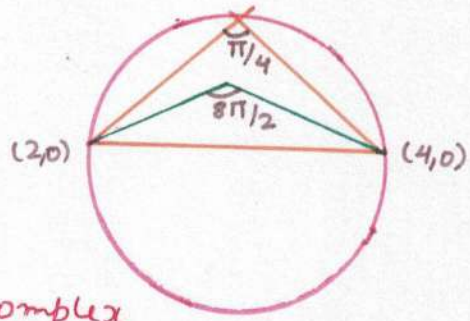
$$\Rightarrow \boxed{x^2 + y^2 - 6x + 8 = 0}$$

(b) $\tan^{-1}\left(\frac{y}{x-4}\right) - \tan^{-1}\left(\frac{y}{x-2}\right) = \frac{\pi}{4}$

$$\tan^{-1}\left(\frac{y(x-2) - y(x-4)}{(x-4)(x-2) + y^2}\right) = \tan^{-1}(1)$$

$xy - 2y - xy + 4y = x^2 + y^2 - 6x + 8$

$x^2 + y^2 - 6x - 2y + 8 = 0$



Ques: $z^2 + pz + q = 0$ is satisfied by 2 complex no.'s and angle z_1 & z_2 is α and $|z_1| = |z_2|$, then prove that - $p^2 = 4q\cos^2 \alpha/2$

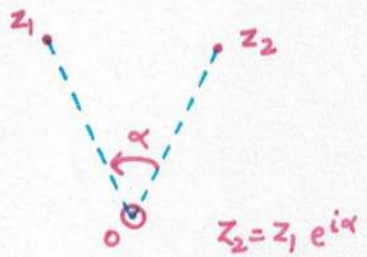
Solu: $-p = z_1 + z_2 \qquad z_1 z_2 = q$

$(z_1 - z_2)^2 = (z_1 + z_2)^2 - 4z_1 z_2$

$= p^2 - 4q$

$4q + (z_1 - z_1 e^{i\alpha})^2 = p^2$

$4q + z_1^2 (1 - e^{i\alpha}) = p^2$



Also, $2z = -p \pm \sqrt{p^2 - 4q}$

$2z_1 = -p + \sqrt{p^2 - 4q}$

$2z_2 = -p - \sqrt{p^2 - 4q}$

DE MOIVRE'S THEOREM

$\text{Cis}\theta = (\cos\theta + i\sin\theta)^n \rightarrow \cos(n\theta) + i\sin(n\theta)$

$n \in Z \rightarrow$ only 1 value

$\rightarrow \cos\left(\frac{2k\pi + p\theta}{q}\right) + i\sin\left(\frac{2k\pi + p\theta}{q}\right)$

$n \in Q \rightarrow$ There will be q values

$\rightarrow \cos(2k\pi + \theta)n + i\sin(2k\pi + \theta)n$

$n \in Q^c \rightarrow$ there will be ∞ values

Ques: Find the roots of $x = (1)^{1/7}$

Solu: $x^7 - 1 = 0$

It should have 7 roots [Be it real or complex]

$x = (1)^{1/7}$

can be written as

$Z = r(\cos\theta + i\sin\theta)^{1/7}$

Where $\theta = \tan^{-1}\left(\frac{0}{1}\right) = 0$

$r = 1$

$1/7$ is rational $\Rightarrow$

$$(\cos\theta + i\sin\theta)^{1/7} = \cos\left(\frac{2k\pi + p\theta}{q}\right) + i\sin\left(\frac{2k\pi + p\theta}{q}\right)$$

$p = 1,\ q = 7,\ \theta = 0$

$K = 0$ 1st root $\rightarrow$ 1

$K = 1$ 2nd root $\rightarrow$ $\cos\frac{2\pi}{7} + i\sin\frac{2\pi}{7}$

$K = 2$ 3rd root $\longrightarrow$ $\cos\frac{4\pi}{7} + i\sin\frac{4\pi}{7}$

⋮

$K = 7$ 7th root $\longrightarrow$ $\cos\frac{12\pi}{7} + i\sin\frac{12\pi}{7}$

Ques: Find the root of $z^4 = 1 + \sqrt{3}i$

Solu: $r = 2,\quad \theta = \pi/3$

$$z^4 = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$z = 2^{1/4}\underbrace{\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^{1/4}}_{\text{4 values}}$$

$$z = 2^{1/4}\left[\cos\left(\frac{2k\pi + \pi/3}{4}\right) + i\sin\left(\frac{2k\pi + \pi/3}{4}\right)\right]$$

$K = 0$ 1st root $\rightarrow$ $z = 2^{1/4}\left[\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right]$

$K = 1$ 2nd root $\rightarrow$ $z = 2^{1/4}\left[\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\right]$

K=2 Third root → $z = 2^{1/4}\left[\cos\frac{13\pi}{12} + i\sin\frac{13\pi}{12}\right]$

K=3 4th root → $z = 2^{1/4}\left[\cos\frac{19\pi}{12} + i\sin\frac{19\pi}{12}\right]$

Ques: Find z satisfying, $z^6 = \frac{1+5i}{3i-2}$

Solu: $z^6 = \frac{(1+5i)}{-2+3i} \times \frac{(-2-3i)}{(-2-3i)}$

$= \frac{(-2+15) + i(-3-10)}{4+9}$

$= \frac{13-13i}{13} = 1-i$

$z = (1-i)^{1/6}$

$z = \sqrt{2}^{1/6}\left(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4}\right)^{1/6}$

6 values

$z = \sqrt{2}^{1/6}\left[\cos\left(\frac{2K\pi + 7\pi/4}{6}\right) + i\sin\left(\frac{2K\pi + 7\pi/4}{6}\right)\right]$

Roots → $z = \sqrt{2}^{1/6}\,\mathrm{cis}\left(\frac{7\pi}{4}\right)$, $\sqrt{2}^{1/6}\,\mathrm{cis}\left(\frac{15\pi}{24}\right)$,

$\sqrt{2}^{1/6}\,\mathrm{cis}\left(\frac{23\pi}{24}\right)$, $\sqrt{2}^{1/6}\,\mathrm{cis}\left(\frac{31\pi}{24}\right)$, $\sqrt{2}^{1/6}\,\mathrm{cis}\left(\frac{39\pi}{24}\right)$

$\sqrt{2}^{1/6}\,\mathrm{cis}\left(\frac{47\pi}{24}\right)$

CUBE ROOT OF UNITY

$z^3 = 1 \Rightarrow z = (1)^{1/3}$ $\quad r = 1, \theta = 0$

$z = \cos\frac{2K\pi}{3} + i\sin\frac{2K\pi}{3}$

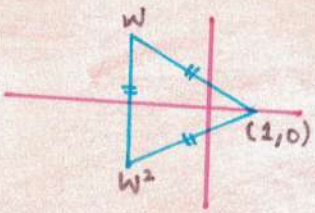
Roots → $1, \quad \frac{-1+\sqrt{3}i}{2} \hookrightarrow w \quad , \quad \frac{-1-\sqrt{3}i}{2} \hookrightarrow w^2$

$z^3 - 1 = 0 \Leftarrow \begin{matrix} 1 \\ w \\ w^2 \end{matrix} \Rightarrow$ $1 + w + w^2 = 0$

$w + w^2 + w^3 = 0$

$w^3 = 1$

NOTES



i) 1, w, and w^2 form an eq. Δ

ii) $1 + x + x^2 = (x-w)(x-w^2)$

$1 - x + x^2 = (x+w)(x+w^2)$

iii) The roots of cube root of unity are diffred by an angle $2\pi/3$

iv) $1 + w^r + w^{2r} = \begin{cases} 3 & \text{If } r \text{ is multiple of } 3 \\ 0 & \text{otherwise} \end{cases}$

v) $\bar{w} = w^2$

vi) $w^{3k+1} = w \quad , \quad w^{3k+2} = w^2 \qquad [k \in I]$

vii) $a^3 - b^3 = (a-b)(a-bw)(a-bw^2)$

viii) $a^3 + b^3 = (a+b)(a+bw)(a+bw^2)$

ix) $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a+bw+cw^2)(a+bw^2+cw)$

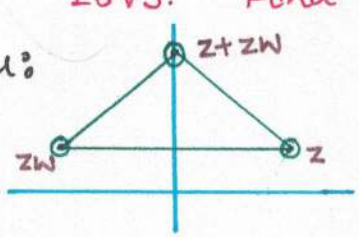
if a, b, c are sides of Δ, then

$a + bw + cw^2 = 0$

⇒ the Δ is equilateral Δ

Ques: Area of Δ formula by z, zw and z + zw is $16\sqrt{3}$. Find |z| ?

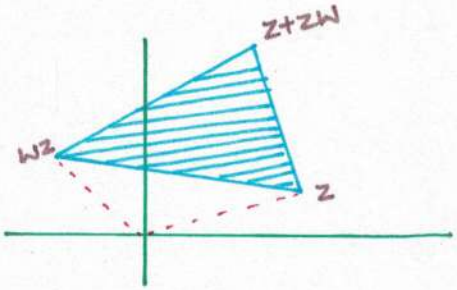
Solu:



$16\sqrt{3} = \frac{\sqrt{3}}{4} (\text{side})^2$

$16 \times 4 = (\text{side})^2$

$\text{side} = 8$



$$w = \frac{-1+\sqrt{3}i}{2}$$

$$z = \left(x-\frac{1}{2}\right) + i\left(y+\frac{\sqrt{3}}{2}\right)$$

∵ The Δ is a equilateral, sides are z, zw, $z(1+w)$ and

$$z+zw+z(1+w) = z(1+w-w^2) = 0$$

$$\boxed{|z| = 8}$$

Ques:- If $(a+w)^{-1} + (b+w)^{-1} + (c+w)^{-1} + (d+w)^{-1} = 2w^{-1}$

and $(a+w^2)^{-1} + (b+w^2)^{-1} + (c+w^2)^{-1} + (d+w^2)^{-1} = 2w^{-2}$

Then prove that: (i) $\sum abc = 2$

(ii) $\sum a = 2\pi a$

(iii) $(a+1)^{-1} + (b+1)^{-1} + (c+1)^{-1} + (d+1)^{-1} = 2$

Solu:- (i) $abc + bcd + cda + dab = 2$

Equation - $\frac{1}{a+x} + \frac{1}{b+x} + \frac{1}{c+x} + \frac{1}{d+x} - \frac{2}{x} = 0$

A biquadratic equation with roots w and w^2

$$(b+x)(c+x)(d+x)(x) + (a+x)(c+x)(d+x)x +$$
$$(a+x)(b+x)(d+x)x + (a+x)(b+x)(c+x)x$$
$$- 2(a+x)(b+x)(c+x)(d+x) = 0$$

METHOD OF FINDING THE SQUARE ROOT OF A COMPLEX NO.

$$\sqrt{a+ib} = x+iy$$

$$a+ib = x^2 - y^2 + 2ixy$$

$$x^2 - y^2 = a \quad , \quad 2xy = b$$

$$(x^2+y^2)^2 = a^2+b^2$$

$$x^2+y^2 = \sqrt{a^2+b^2}$$

Direct formula for square root of a complex no.

$$(x+iy)^{1/2} = \pm\left(\sqrt{\frac{|z|+Re(z)}{2}} \pm \sqrt{\frac{|z|-Re(z)}{2}}\right)$$

If $Im(z) > 0$, use (+) in between, if $Im(z) < 0$, use (−)

Ques: Find the square root of $-7-24i$

Solu: $(x^2+y^2) = 49 + 576 = 625$

$x^2+y^2 = 25 \qquad x^2-y^2 = -7$

$2x^2 = 18 \qquad 2xy = -24$

$x = \pm 3 \qquad y = \pm 4$

$$\sqrt{z} = 3-4i \quad \text{or} \quad -3+4i$$

Ques: $9+40i$

Solu: $81 + 1600 = 1681$

$x^2+y^2 = 41$

$x^2-y^2 = 9$

$2x^2 = 50$

$x = \pm 5$

$x^2-y^2 = 9$

$25-9 = y^2$

$y = \pm 4$



Root → $5+4i, -5-4i$

Ques: $(-81)^{1/4}$

Solu: $z = (-81)^{1/4} \Rightarrow z^2 = (-81)^{1/2}$

$-81 = x+iy$

$z^2 = (+9i)^{2\times 1/2}$

$z^2 = 9i$

$z = \sqrt{9i}$

$x^2-y^2 = 0$

$x^2+y^2 = 9$

$x = \pm 3/\sqrt{2}$

$x^2-y^2 = -81$

$2xy = 0$

$x^2 = -81$

Root → $\frac{3}{\sqrt{2}}(1 \pm i)$

Chapter- Quadratic Equations
Class-11th

Identities :-

which is true for every value permissible of 'x'
which has more roots then its degree.
In which coefficient of each term is zero.

Note :- Degree :- highest power of x present.

Ex → $\boxed{(x+3)^2 = x^2+6x+9}$ → Identities

Ques Find value of 'k' for which $(k^2-4)x^2+(k^2-5x+6)x+k^2+5k-14=0$ an identity in x.

Ans

$$\downarrow$$

$$0x^2+0x+0=0$$

$k^2-4=0 \Rightarrow k=\pm 2$

$k^2-5k+6=0 \Rightarrow k=2,3$

$k^2+5k-14=0 \Rightarrow k=-7,2$

$\} \wedge k = 2$

Polynomial Equations :-

	degree
# Linear Equations :- $ax+b=0,\ a\neq 0$	1
# Quadratic Equations $ax^2+bx+c=0,\ a\neq 0$	2
# Cubic Equation $ax^3+bx^2+cx+d=0,\ a\neq 0$	3
# Bi-quadratic Equations $ax^4+bx^3+cx^2+dx+e=0,\ a\neq 0$	4

Quadratic Equations :-

$ax^2+bx+c=0$

where :- $a, b, c \in R,\ a\neq 0$

$ax^2+bx+c=0$

Perfect square :-

$$a\left(x^2+\frac{bx}{a}+\frac{c}{a}\right)=0$$

$$x^2+\frac{2bx}{2a}+\frac{b^2}{4a^2}-\frac{b^2}{4a}+\frac{c}{a}=0$$

$$\left(x+\frac{b}{2a}\right)^2=\frac{b^2}{4a}-\frac{c}{a}$$

$$\left(x+\frac{b}{2a}\right)^2=\frac{b^2-4ac}{4a^2}$$

$$x+\frac{b}{2a}=\pm\sqrt{\frac{b^2-4ac}{4a^2}}$$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$ax^2+bx+c=0$

$$\alpha=\frac{-b-\sqrt{b^2-4ac}}{2a}$$

$$\beta=\frac{-b+\sqrt{b^2-4ac}}{2a}$$

$$b^2-4ac=D$$

Discriminant

Sum of roots $=\alpha+\beta=\frac{-b}{a}$

Product of roots $=\alpha\beta=\frac{c}{a}$

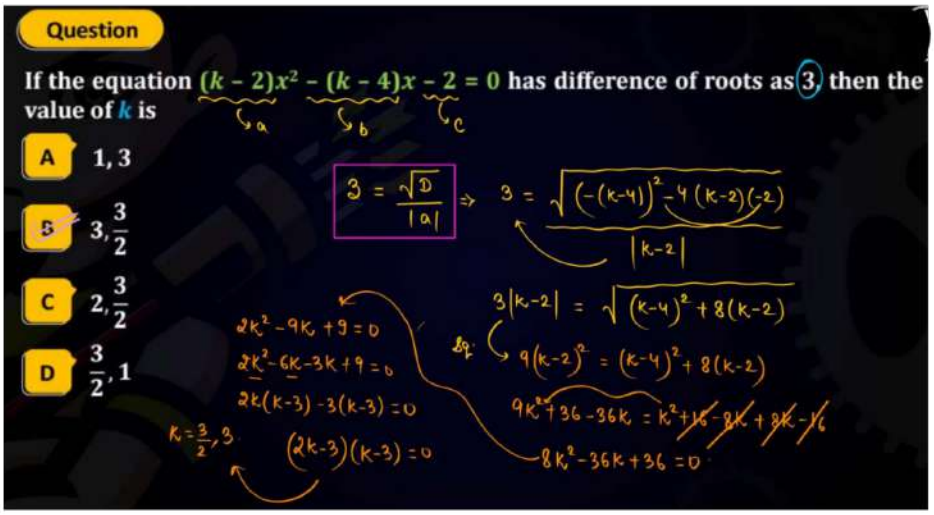
Difference of roots $=|\alpha-\beta|=\frac{\sqrt{D}}{|a|}$

Sum, Product & Difference of roots :-

(I) Sum of roots $=\alpha+\beta=\frac{-b}{a}=-\frac{(\text{coefficient of } x)}{(\text{coefficient of } x^2)}$

(II) Product of roots $=\alpha\beta=\frac{c}{a}=\frac{\text{constant term}}{\text{coefficient of } x^2}$

(III) Difference of roots $=|\alpha-\beta|=\frac{\sqrt{D}}{|a|}$



Important Notes :-

Note :- 1 Factorise of Quadratic :-

ek quadratic ko factorise krke, equate kr diya jaye to vo

identity bn jati hai.

This method is valid for every polynomial equation.

Note:-2 Forming Quadratic if Roots are given:-

Example

If α and β be roots of the equation $2x^2 - 3x - 7 = 0$. Without computing α and β, set a quadratic equation with α/β and β/α as its roots.

$2x^2 - 3x - 7 = 0$ (α, β)

$\alpha + \beta = -\frac{(-3)}{2} = \frac{3}{2}$

$\alpha\beta = \frac{-7}{2}$

$\alpha^2 + \beta^2 + 2\alpha\beta = \frac{9}{4}$

$\alpha^2 + \beta^2 = \frac{9}{4} - 2\left(\frac{-7}{2}\right) = \frac{9}{4} + 7 = \frac{9+28}{4} = \frac{37}{4}$

$x^2 - \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)x + \left(\frac{\alpha}{\beta}\cdot\frac{\beta}{\alpha}\right) = 0$ ($\frac{\alpha}{\beta}$, $\frac{\beta}{\alpha}$)

$x^2 - \left(\frac{\alpha^2+\beta^2}{\alpha\beta}\right)x + 1 = 0$

$x^2 - \left(\frac{37}{4} \div \frac{-7}{2}\right)x + 1 = 0$

$x^2 + \frac{37}{14}x + 1 = 0$

$14x^2 + 37x + 14 = 0$

Note 3:- i) If α & β are root of any quadratic then they will satisfy that equatic

ii) Any higher degree polynomial expression in α & β canbe reduced to linear or constant

Question

If $\alpha^2 - 13 = 5\alpha$ and $\beta^2 = 13 + 5\beta$, then find value of

(i) αβ

(ii) $\alpha^2\beta + \alpha\beta^2 + 7\alpha + 7\beta + 2$

$\alpha^2 - 5\alpha - 13 = 0$

$\beta^2 - 5\beta - 13 = 0$

$\Rightarrow x^2 - 5x - 13 = 0$ (α, β)

$\alpha\beta = -13$

$\alpha + \beta = -\frac{(-5)}{1} = 5$

$\alpha\beta(\alpha+\beta) + 7(\alpha+\beta) + 2$

$-13\left(\frac{5}{1}\right) + 7(5) + 2$

$-65 + 35 + 2 = -30 + 2$

$= -28$

Note:- 4 Observation

Quadratic: $ax^2 + bx + c = 0$

i) If $a+b+c = 0$ then root of Quadratic are 1 & $\frac{c}{a}$

ii) if $a-b+c = 0$ then root of Quadratic are -1 & $-\frac{c}{a}$

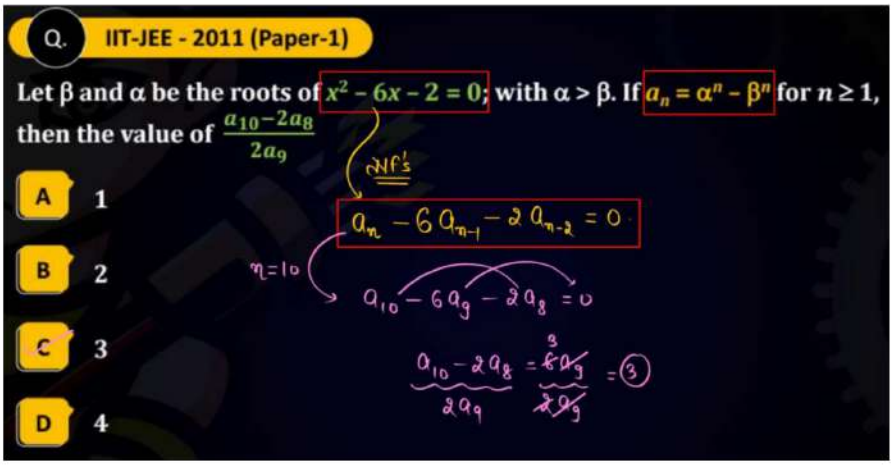
iii) if $a-2b+4c = 0$ then root of Quadratic are $-\frac{1}{2}$ & $-\frac{2c}{a}$

Newton's Formula:-

$ax^2 + bx + c = 0$ → α, β } are roots

if we define $S_n = \alpha^n \pm \beta^n$ / $S_n = p\alpha^n \pm q\beta^n$

Results $\boxed{aS_n + bS_{n-1} + cS_{n-2} = 0}$



Nature of roots :-

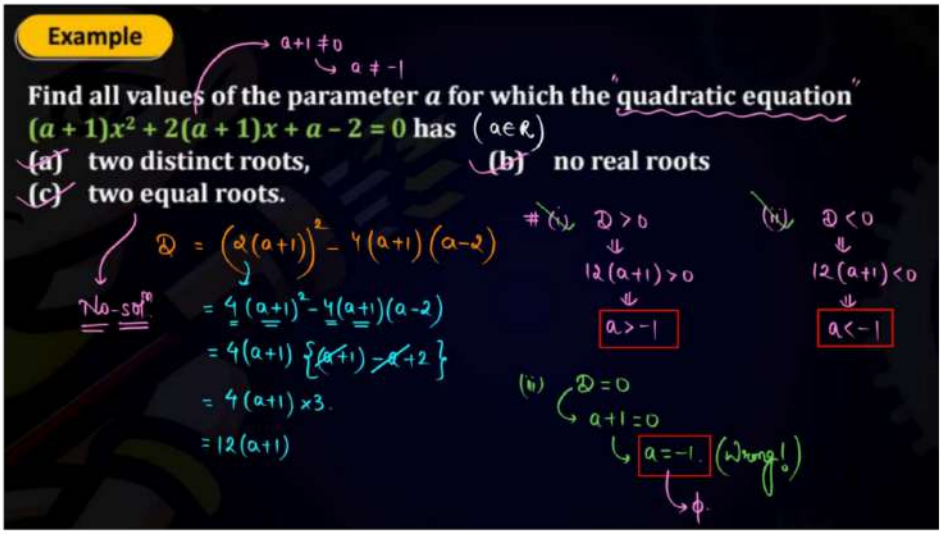
$$ax^2 + bx + c = 0, \quad a, b, c \in R, \ a \neq 0$$

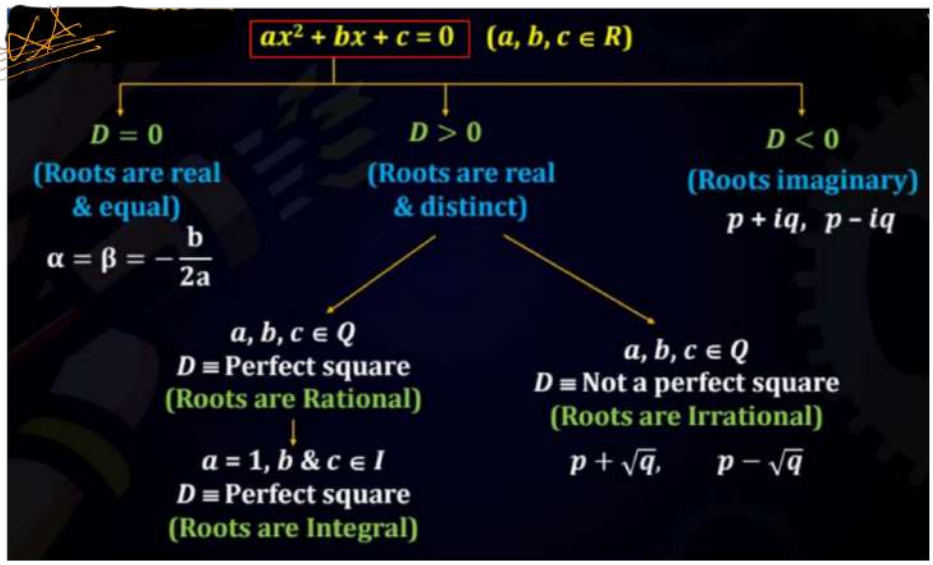
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \longrightarrow D$$

a) If $D > 0$ then Real & distinct root

b) If $D = 0$ then Real & Equal root

c) If $D < 0$ then Root are imaginary.





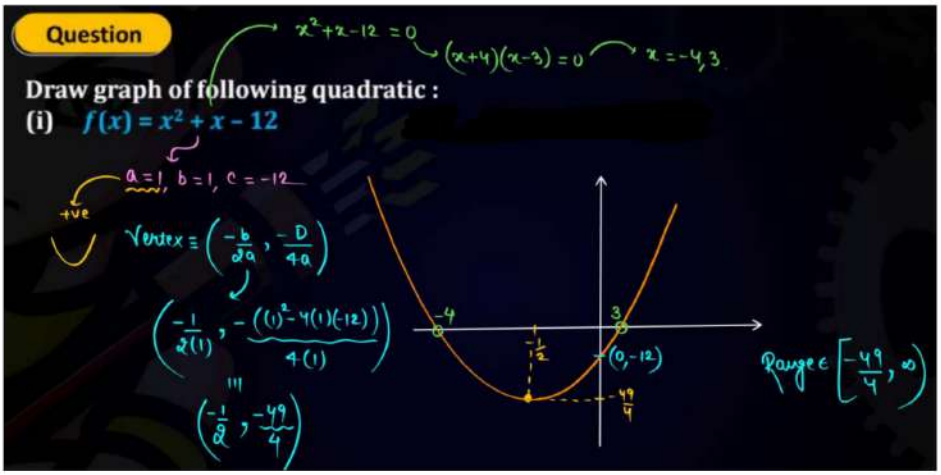
Graph of Quadratic Equations :-

Consider three things :-

i) coefficient of $x^2 = a$

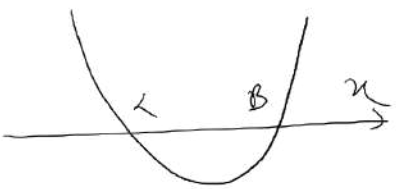
ii) vertex $= \left(\frac{-b}{2a}, \frac{-D}{4a}\right)$

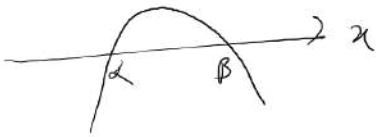
iii) point where it's cuts Y-axis is $(0, c)$



Graph with respect to Discriments :-

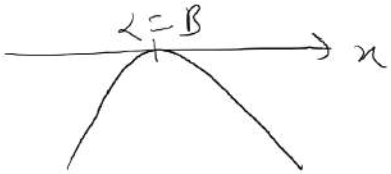
i) If $D > 0$





ii) If $D = 0$

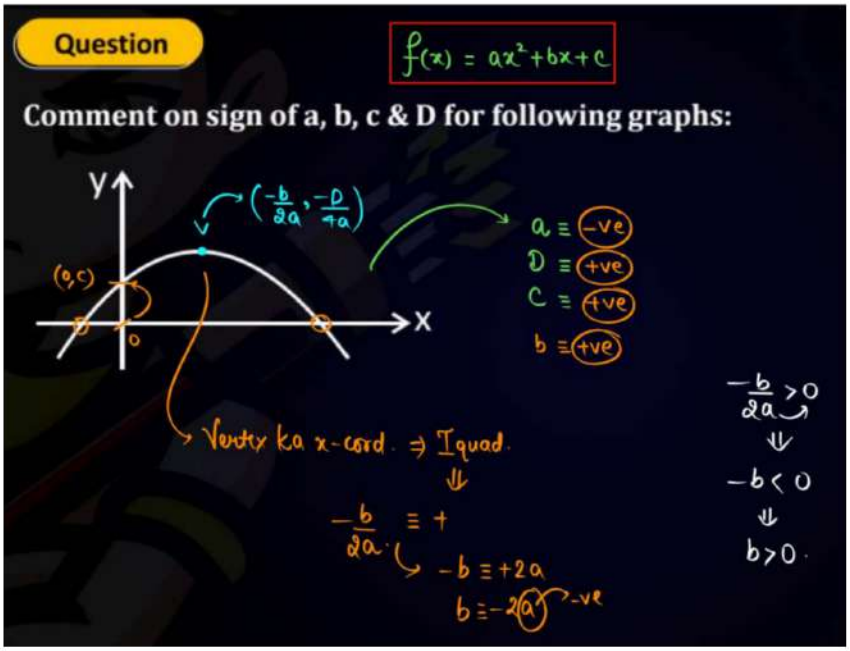




iii) If $D < 0$







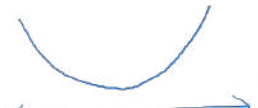
Range of Quadratic

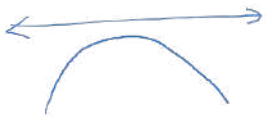
If $f(x) = ax^2 + bx + c$

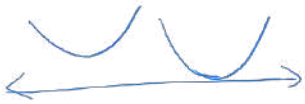
then if $a > 0$ then Range $\in \left[\frac{-D}{4a}, \infty\right)$

then if $a < 0$ then Range $\in \left(-\infty, \frac{-D}{4a}\right]$

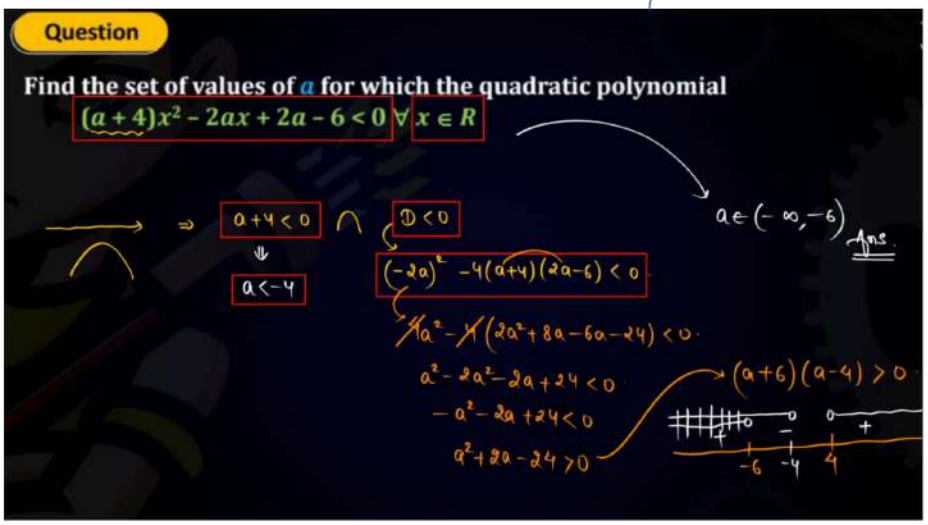
Important Note:-

a) If $ax^2 + bx + c > 0 \ \forall x \in R \iff$  $\iff a > 0 \cap D < 0$

b) If $ax^2 + bx + c < 0 \ \forall x \in R \iff$  $\iff a < 0 \cap D < 0$

c) If $ax^2 + bx + c \geq 0 \ \forall x \in R \iff$  $\iff a > 0 \cap D \leq 0$

d) If $ax^2 + bx + c \leq 0 \ \forall x \in R \iff$  $\iff a < 0 \cap D \leq 0$



Range!-

format!- Quad/Quad, Linear/Quad, Quad/Linear, Constant/Quad

Step to be followed:-

I) Assume $y =$ ---(1)
cross-multiply & form a Quadratic in x.

II) Make $D \geq 0$ or above quadratic (as $x \in$ Real)

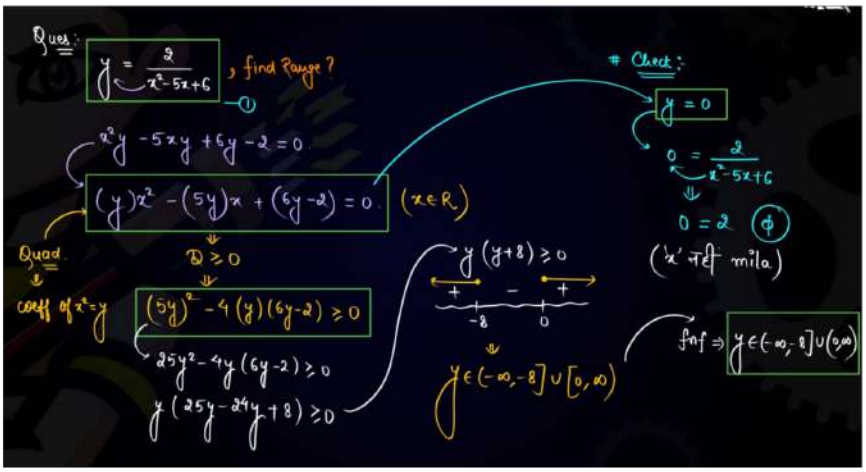
III) Solver above inequaily using wavy curve.

IV) Check your answer

Make coefficient of $x^2 = 0$
obtain 'y' & put the value of y in equation (1)

If you get any x - Then S-(III) final ans

If you don't get any x → Then remove that 'y' from III.



Common Root condition!-

Case-I : If Both roots are common!-

$a_1x^2 + b_1x + c_1 = 0$ → α, β

$a_2x^2 + b_2x + c_2 = 0$ → α, β

Condition $\boxed{\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}}$

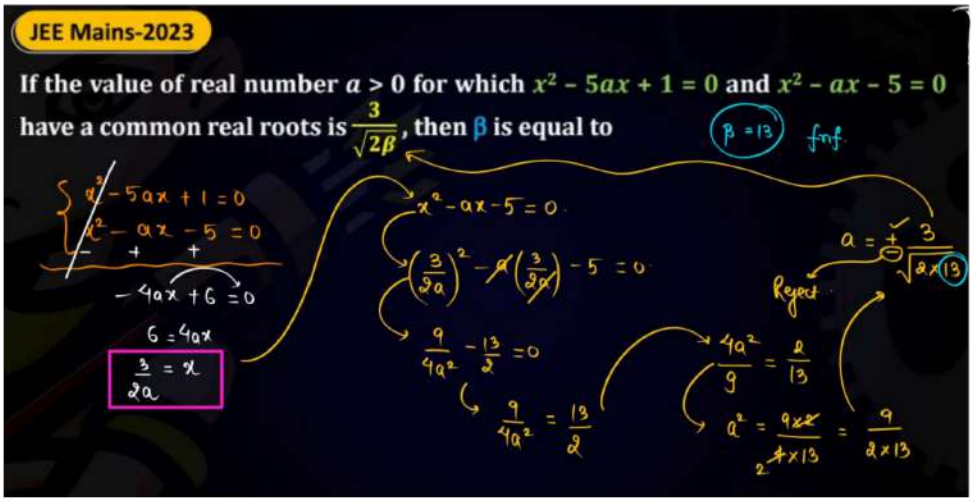
Case-II : If atleast one root is common :-

$a_1x^2 + b_1x + c_1 = 0$ → α, β

$a_2x^2 + b_2x + c_2 = 0$ → α, γ

Process :- i) make coefficient of x^2 same then subtract both equation and get $(x = \alpha)$

ii) $(x = \alpha)$ is common root & it will satisfy both equations.

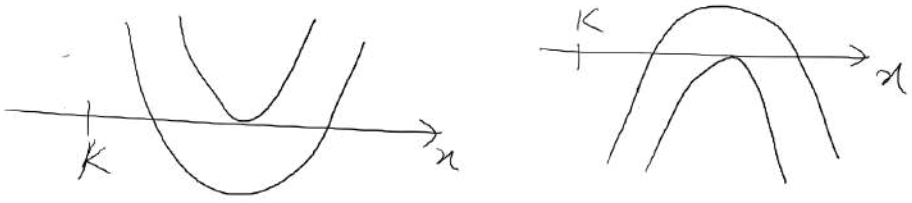


Location of roots :-

Condition on :- a, D, x-coordinate of vertex, value of of function at $x = k$.

Case-I If both roots are greater than number given number 'k'.

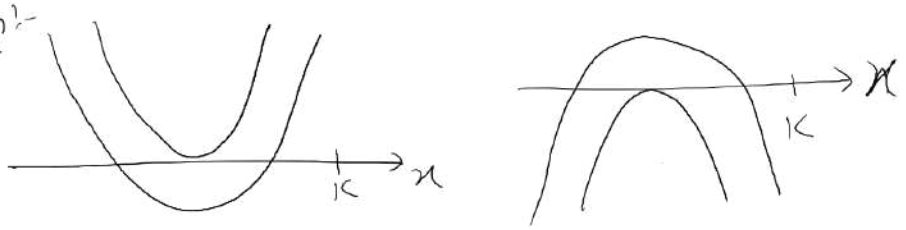
Possible graph :-



Condition :- $\boxed{D \geq 0 \wedge \frac{-b}{2a} > k \wedge a \cdot f(k) > 0}$

Case-II :- If both root are less than number given number 'k'.

Possible graph :-



Condition :- $\boxed{D \geq 0 \wedge \frac{-b}{2a} < k \wedge a \cdot f(k) > 0}$

Question

Find all the values of the parameter d for which both roots of the equation $x^2 - 6dx + (2 - 2d + 9d^2) = 0$ exceed the number 3

$f(x)$

greater than $3 = k$

$D \geq 0$

$(-6d)^2 - 4(1)(2 - 2d + 9d^2) \geq 0$

$36d^2 - 8 + 8d - 36d^2 \geq 0$

$8(d-1) \geq 0$

$d \geq 1$

$\frac{-b}{2a} > 3$

$\frac{3(-6d)}{2(1)} > 3$

$3d > 3$

$d > 1$

$af(3) > 0$

$1 \cdot ((9 - 18d) + 2 - 2d + 9d^2) > 0$

$(9d^2 - 20d + 11) > 0$

$9d^2 - 11d - 9d + 11 > 0$

$d(9d-11) - 1(9d-11) > 0$

$(d-1)(9d-11) > 0$

$d = 1, \; d = \frac{11}{9}$

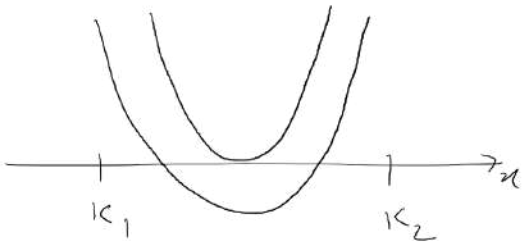


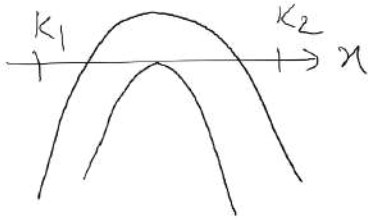
$d \in \left(\frac{11}{9}, \infty\right)$

fnf

Case-III :- Both root lies between 'k_1' & 'k_2' : I ∧ II

Possible garph



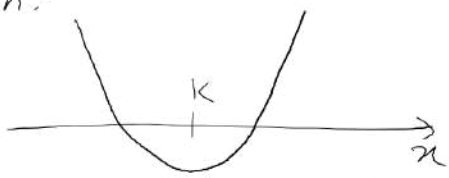


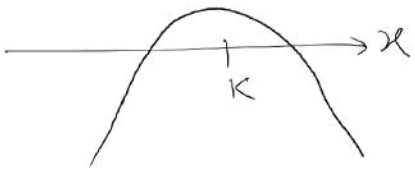
condition :-

$$D \geq 0 \; \wedge \; k_1 < \frac{-b}{2a} < k_2 \; \wedge \; a \cdot f(k_1) > 0 \; \wedge \; af(k_2) > 0$$

Case-IV :- Any number K lies between the roots :-

Possible garph :-



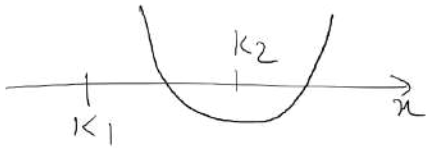


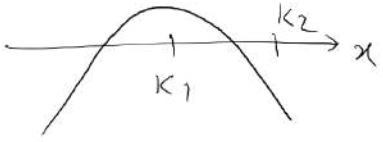
Condition

$$af(k) < 0$$

Case-V :- If exactly one root of quadratic lies between k_1 & k_2

Possible graph :-



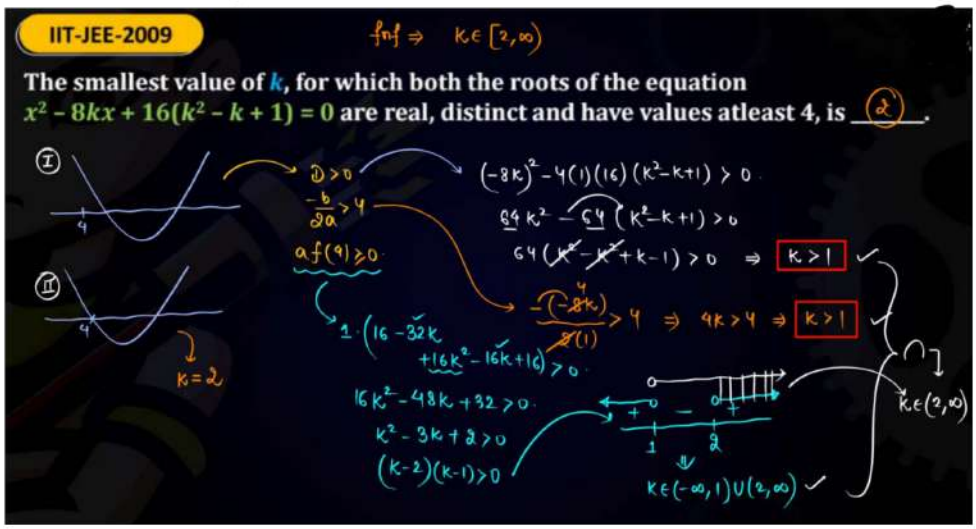


conditions :-

$$f(k_1)\, f(k_2) < 0$$

Important Note:-

Boundary condition should be separately analysed.



Theory of higher Degree Equations:-

Cubic Equation:-

$$ax^3 + bx^2 + cx + d = 0, \quad a \neq 0$$

Sum of roots $= \alpha+\beta+\gamma = \frac{-b}{a}$

Sum of product of root taken two at a time $= \alpha\beta+\beta\gamma+\gamma\alpha = \frac{c}{a}$

Product of roots $= \alpha\beta\gamma = \frac{-d}{a}$

Important Note:-

I) Every odd degree polynomial equation with real coefficient always have 1 real root. (Imaginary root exists in Pair)

II) Newton's formula : $S_n = \alpha^n \pm \beta^n \pm \gamma^n$

Then $aS_n + bS_{n-1} + cS_{n-2} + dS_{n-3} = 0$

III) $ax^3 + bx^2 + cx + d = a(x-\alpha)(x-\beta)(x-\gamma)$

IV) form a cubic whose roots are α, β, γ.

$$x^3 - S_1x^2 + S_2x - S_3 = 0$$

$S_1 \to \alpha+\beta+\gamma$, $S_2 \to \alpha\beta+\beta\gamma+\gamma\alpha$, $S_3 \to \alpha\beta\gamma$

Quartic Equation :-

$$ax^4 + bx^3 + cx^2 + dx + e = 0$$

S_1 = Sum of root = $\Sigma\alpha = \alpha+\beta+\gamma+\delta = -\frac{b}{a}$

S_2 = Sum of product of roots taken two at a time = $\Sigma\alpha\beta = \frac{c}{a}$

$= \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \frac{c}{a}$

S_3 = Sum of product of roots taken three at a time = $\Sigma\alpha\beta\gamma$

$= \alpha\beta\gamma + \beta\gamma\delta + \alpha\beta\delta + \alpha\gamma\delta = -\frac{d}{a}$

S_4 = product of roots = $\alpha\beta\gamma\delta = \frac{e}{a}$

Equations Reducible To Quadratic :-

format - I $\boxed{(x-a)(x-b)(x-c)(x-d) = K}$

aise multiply ( )( ) such that variable part same one.

Ex →

$$(x+1)(x+2)(x+3)(x+4) = \frac{9}{16}$$

$$(x+1)(x+4)(x+2)(x+3) = \frac{9}{16}$$

$$(x^2+5x+4)(x^2+5x+6) = \frac{9}{16}$$

Let $x^2+5x+4 = t$

$$t(t+2) = \frac{9}{16}$$

$$16t^2 + 32t = 9$$

$$16t^2 + 32t - 9 = 0$$

$$(4t+9)(4t-1) = 0$$

$$t = +\frac{1}{4}, -\frac{9}{4}$$

format - II $\boxed{(x-a)(x-b)(x-c)(x-d) = Kx^2}$

aise multiply ( ) ( ) karo ki constant part same.

Ex - $(x+2)(x+3)(x+8)(x+12) = 4x^2$

$$(x+2)(x+12)(x+3)(x+8) = 4x^2$$

$$\left(\frac{x^2+14x+24}{x}\right)\left(\frac{x^2+11x+24}{x}\right) = \frac{4x^2}{x^2}$$

$$\left(x+14+\frac{24}{x}\right)\left(x+11+\frac{24}{x}\right) = 4$$

Let $x + 11 + \frac{24}{x} = t$

$$(t+3)t = 4$$

format - III

$(x-a)^4 + (x-b)^4 = K$

↘ Put $\frac{(x-a)+(x-b)}{2} = t$

$x = t + \left(\frac{a+b}{2}\right)$

format - IV

Reciprocal Equation :-

$ax^3 + bx^2 + bx + a = 0$

$ax^4 + bx^3 + cx^2 + bx + a = 0$

$ax^5 + bx^4 + cx^3 + cx^2 + bx + a = 0$ → Hint one root $x = \pm 1$

$ax^3 + bx^2 + bx + a = 0$

$a(x^3+1) + bx(x+1) = 0$

$a(x+1)(x^2-x+1) + bx(x+1) = 0$

$(x+1)\{a(x^2-x+1) + bx\} = 0$

$ax^4 + bx^3 + cx^2 + bx + a = 0$

divide whole equation by x^2

$ax^2 + bx + c + \frac{b}{x} + \frac{a}{x^2} = 0$

$a\left(x^2 + \frac{1}{x^2}\right) + b\left(x + \frac{1}{x}\right) + c = 0$ → Say $x \pm \frac{1}{x} = t$

